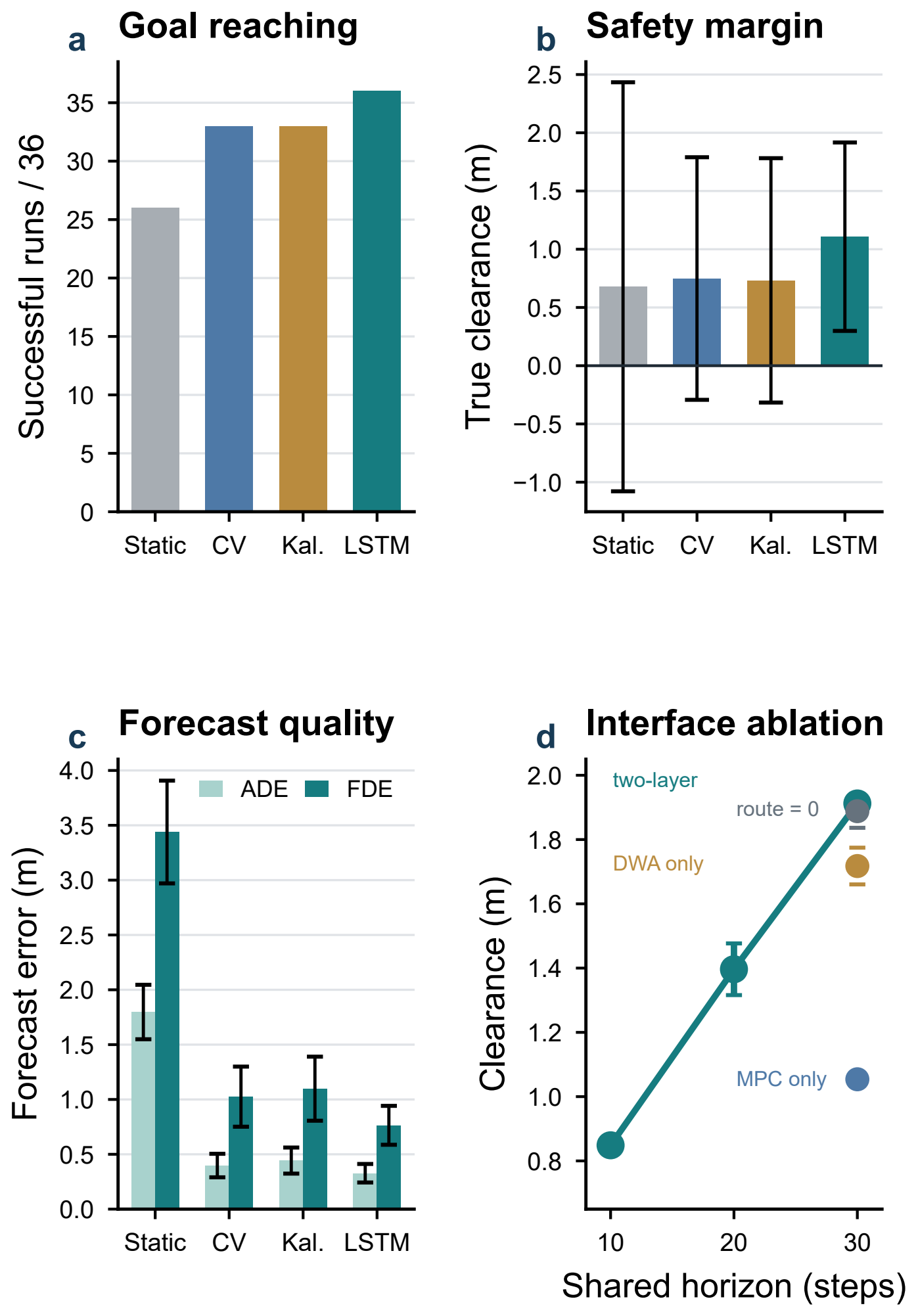
a
Goal reaching
Successful runs / 36
0
5
10
15
20
25
30
35
Static
CV
Kal.
LSTM
b
Safety margin
True clearance (m)
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
Static
CV
Kal.
LSTM
c
Forecast quality
Forecast error (m)
ADE
FDE
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Static
CV
Kal.
LSTM
d
Interface ablation
Clearance (m)
0.8
1.0
1.2
1.4
1.6
1.8
2.0
two-layer
route = 0
DWA only
MPC only
10
20
30
Shared horizon (steps)